

Draft - expert review pending

This unit is complete and passes the platform's automated checks, but no reviewer has read it yet. Treat its sources and claims with care.

Unit 3 assessment

Unit summary

Air pollution runs the source-pathway-receptor chain through the air, carried by six pollutants that matter: particulate matter, nitrogen dioxide, sulfur dioxide, carbon monoxide, ground-level ozone and lead. The WHO estimates 4.2 million premature deaths a year from outdoor air, and nearly the whole world breathes air above its guidelines; Lahore's inversion-trapped winter smog is the national case, and controls act at the source, weakly in the pathway, and at the receptor, where a teacher's AQI decisions live. Noise pollution harms at a dose: hearing loss that is permanent, plus blood pressure, sleep and learning effects, with decibels logarithmic and the damage zone beginning near 85 dB. HSE studies hazards to people and environment together; hazards sort into five classes, physical, mechanical, biological, chemical and psychological, each controlled differently. Risk is likelihood times severity, ranked on a grid, controlled through a hierarchy that puts removal first and personal protection last, in a cycle of identify, assess, control, monitor that repeats.

Summative assessment

Multiple-choice questions (MCQs)

1. Which pollutant is a colourless, odourless gas that takes up oxygen's place in blood? (*Remember*) a) sulfur dioxide b) carbon monoxide c) ground-level ozone d) PM2.5
2. Ground-level ozone is distinctive because it: (*Understand*) a) is emitted directly by vehicles b) forms when sunlight acts on other pollutants c) only exists indoors d) disappears in winter

3. In the air pollution chain, "school closures on hazardous days" acts at the: *(Apply)* a) source b) pathway c) receptor d) cause
4. Every 10 decibels of increase represents: *(Remember)* a) a doubling of sound energy b) a tenfold increase in sound energy c) ten times the loudness as heard d) no physical change
5. A sound meter reads 70 dB in the corridor and 90 dB at the gate. In energy terms the gate is: *(Apply)* a) about 20 per cent louder b) about twice as loud c) about a hundred times the energy d) the same, since hearing adjusts
6. Which of these is a psychological hazard by this unit's classification? *(Understand)* a) a scorpion in the store room b) unmanageable workload during floods c) a broken window frame d) pesticide spray drift
7. The five hazard classes are: *(Remember)* a) air, water, land, noise, light b) physical, mechanical, biological, chemical, psychological c) source, pathway, receptor, effect, control d) likelihood, severity, control, monitor, review
8. A hazard differs from a risk in that the hazard is: *(Understand)* a) the chance of harm, the risk the potential b) the potential to cause harm, the risk the chance and seriousness of it occurring c) always physical, while risks are not d) the same thing after an incident
9. In the hierarchy of controls, the best level is: *(Remember)* a) personal protective equipment b) training and procedures c) removing the hazard entirely d) warning signs
10. The Karachi construction study found its most common hazards were: *(Apply)* a) machinery failures alone b) excavation, unsafe practices and missing protection c) weather only d) unmeasurable

Restricted-response questions (RRQs)

1. Explain, in two sentences, why a temperature inversion worsens a city's winter smog. *(Understand)*
2. Name two reasons why "the wind will clean the air" is a weak control strategy. *(Analyze)*

3. Name four health impacts of noise pollution, only one of which is hearing loss. *(Remember)*
4. Why is a clear sky not evidence that the air is safe? Use one pollutant in your answer. *(Analyze)*
5. Distinguish a hazard from an incident with one example of each from a school. *(Understand)*
6. Name the five hazard classes and give one flood-camp example of each. *(Understand)*
7. Why does psychological harm belong on a hazard register, by the same standard as a scorpion? *(Analyze)*
8. Define risk and name its two axes. *(Remember)*
9. Why does personal protective equipment sit at the bottom of the hierarchy of controls? *(Analyze)*
10. A loose wire (high likelihood, high severity) and a yearly-used wobbly ladder (low likelihood, high severity): which do you fix first, and why? *(Analyze)*

Extended-response questions (ERQs)

1. A parents' committee says the smog is another country's fault, so nothing should change. Using the chain and the Lahore case, give an honest answer that names what is shared, what is ours, and what a school can still do. *(Analyze)*
2. A school beside a main road is told nothing can be done about the noise. Using the three links and the health evidence, write the reply, ending with the two cheapest actions for this term. *(Evaluate)*
3. Your school becomes a flood relief camp for a month. Write a hazard brief for volunteers: at least four hazards across at least three classes, each with a same-day control. *(Analyze)*
4. Write a risk assessment brief for the school fair: at least four hazards across three classes, grid-scored, each with a control at a named hierarchy level, plus a monitoring plan. *(Evaluate)*
5. Integrative: a school downwind of a brick kiln has dusty classrooms, a noisy road in front and an unlit stair inside. Analyse the full situation, air, noise and physical hazards, through the risk cycle, and present one ranked action list with the reasoning shown. *(Evaluate)*

Answers and marking guidance

MCQ answer key

1. b - carbon monoxide binds haemoglobin, taking the seat oxygen needs.
2. b - ozone is secondary, cooked by sunlight from other pollutants, peaking in hot afternoons.
3. c - closures protect the receptors, the pupils; they do not touch the source.
4. b - decibels are logarithmic: 10 dB is ten times the energy.
5. c - 20 dB is two tenfold steps, about a hundred times the energy.
6. b - workload and disaster stress are psychological hazards; the scorpion is biological, the window mechanical, the spray chemical.
7. b - physical, mechanical, biological, chemical, psychological.
8. b - the hazard is the potential; the risk is chance and seriousness in real conditions.
9. c - removal eliminates exposure for everyone; PPE protects one person, while worn.
10. b - the study named excavation, working-practice variation and missing or unused PPE, with faulty machinery.

RRQ model answers and mark schemes

1. Model: the inversion traps cold air, and the city's emissions, under a warm lid, so pollutants cannot disperse upward and accumulate over the city.
Marks (2): 1 for the trapping mechanism; 1 for the accumulation consequence.
2. Model: first, dispersion only moves the problem, it does not remove the pollutant; second, in winter inversions there is little dispersion to help with.
Marks (2): 1 per reason.
3. Model: hearing loss plus any three of raised blood pressure, stress hormones, disturbed sleep, slowed learning in noisy classrooms. **Marks (4):** 1 per impact, max 1 for hearing loss.
4. Model: PM_{2.5} is invisible particle by particle, so a clear sky can still carry an unhealthy index; the eye is not an instrument. **Marks (2):** 1 for the pollutant; 1 for the measurement point.

5. Model: hazard, the wet floor or broken frame, the potential; incident, the slip or cut, the event. **Marks (2):** 1 per definition with example.
6. Model: physical, heat and standing water; mechanical, a collapsing wall; biological, water-borne disease; chemical, cylinder or bleach storage; psychological, displacement stress. **Marks (5):** 1 per class with an example.
7. Model: the register exists to capture anything with the potential to cause harm; psychological harms meet that standard, are predictable after disasters, and are controllable, like any other class. **Marks (2):** 1 for the same-standard argument; 1 for predictability or controllability.
8. Model: risk is the chance and seriousness of harm occurring in real conditions; the axes are likelihood and severity. **Marks (2):** 1 for the definition; 1 for both axes.
9. Model: PPE protects only one person, only while correctly worn, and fails silently when forgotten; higher levels protect everyone by changing the hazard or the path to it. **Marks (2):** 1 for the one-person/while-worn limits; 1 for the comparison upward.
10. Model: the wire first, high likelihood and high severity is the act-now corner, and the fix is cheap; the ladder is controlled meanwhile by tagging it out of use. **Marks (2):** 1 for the choice with grid reasoning; 1 for interim handling.

ERQ rubrics

1. **Analytic rubric.** *Chain used to separate shared from local (0-3, Analyze) · Evidence: pollutant table or the Lahore study (0-2, Apply) · School actions at the receptor named (0-3, Understand) · Clarity and register (0-2).* Total 10.
2. **Analytic rubric.** *Three links applied to noise (0-3, Analyze) · Health evidence beyond annoyance (0-2, Understand) · Two cheap this-term actions (0-3, Apply) · Clarity (0-2).* Total 10.
3. **Analytic rubric.** *Hazards span three or more classes, correctly named (0-3, Apply) · Controls are same-day and volunteer-achievable (0-3, Apply) · Realism of the camp scenario (0-2, Understand) · Clarity (0-2).* Total 10.
4. **Analytic rubric.** *Cycle complete: identify, score, control, monitor (0-3, Apply) · Hierarchy levels named and matched (0-3, Understand) · Monitoring plan concrete, with a person and a time (0-2, Analyze) · Clarity (0-2).* Total 10.

5. **Analytic rubric.** *All three exposure types identified correctly (0-2, Analyze)*
· *Grid reasoning shown in the ranking (0-3, Analyze)* · *Controls placed in the hierarchy (0-2, Apply)* · *One coherent ranked list, justified (0-2, Evaluate)* · *Clarity (0-1).* Total 10.

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